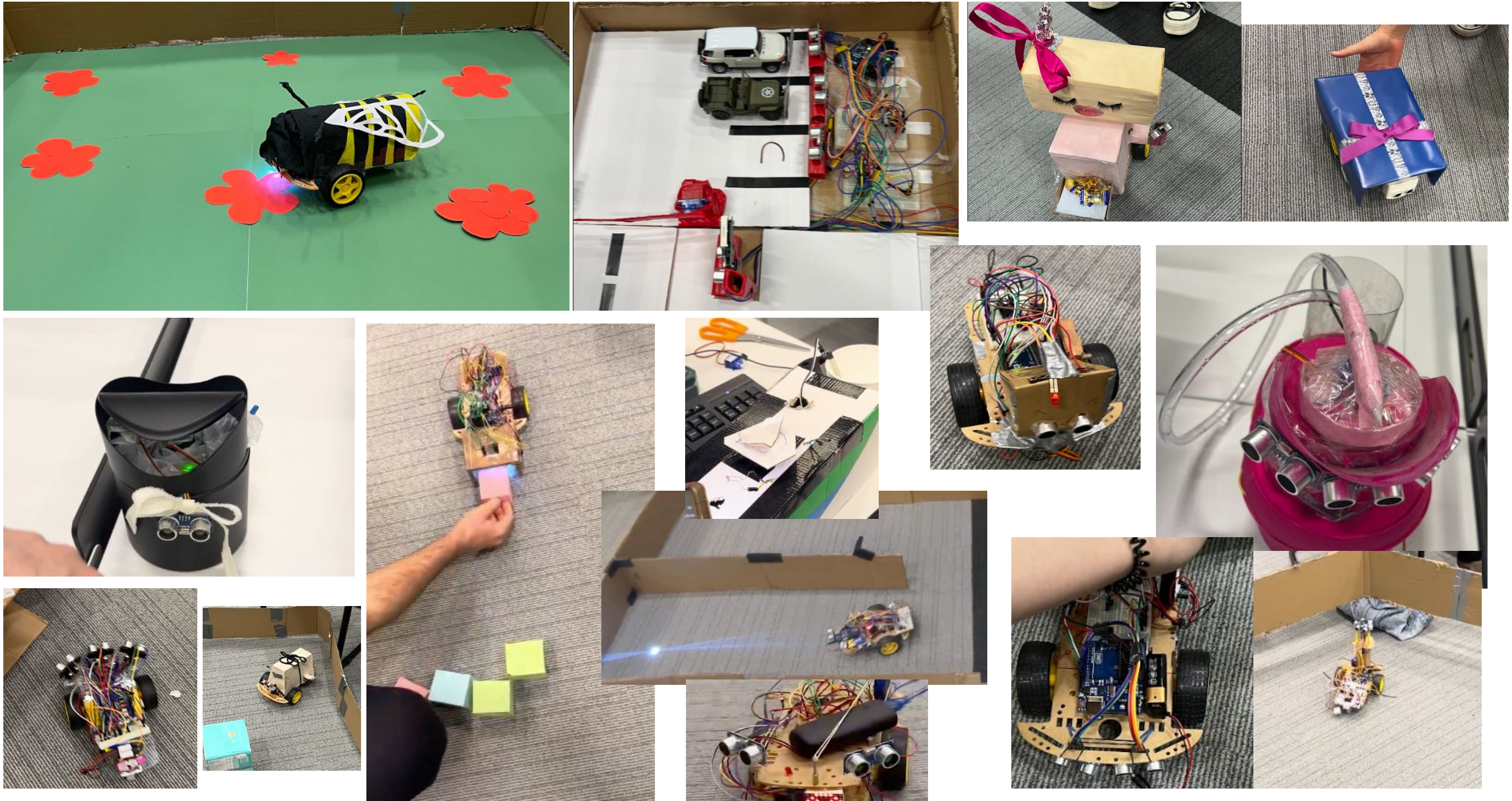
# Autonomous Systems

Week 13: Exam review and final words.

# today: the last lecture

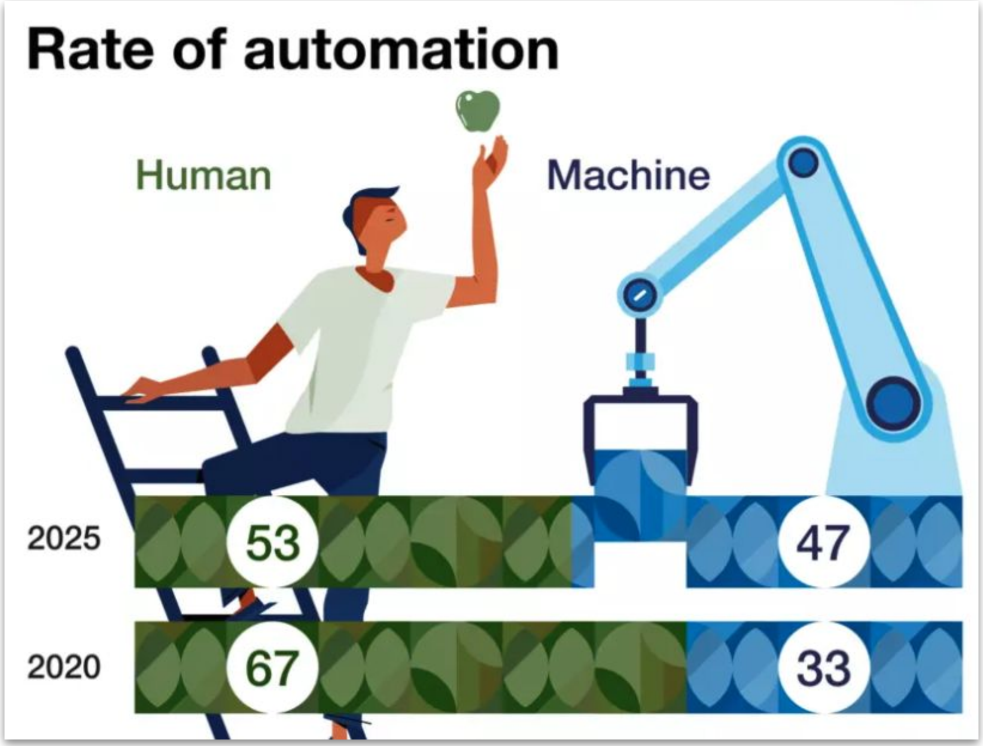
Image generated using dalle, “a sad robot” as prompt.

“ ... **endless forms most beautiful** and most wonderful have been, and are being, evolved.”



# It is a perfect time to be a roboticist


# Remember the first lecture of HCI

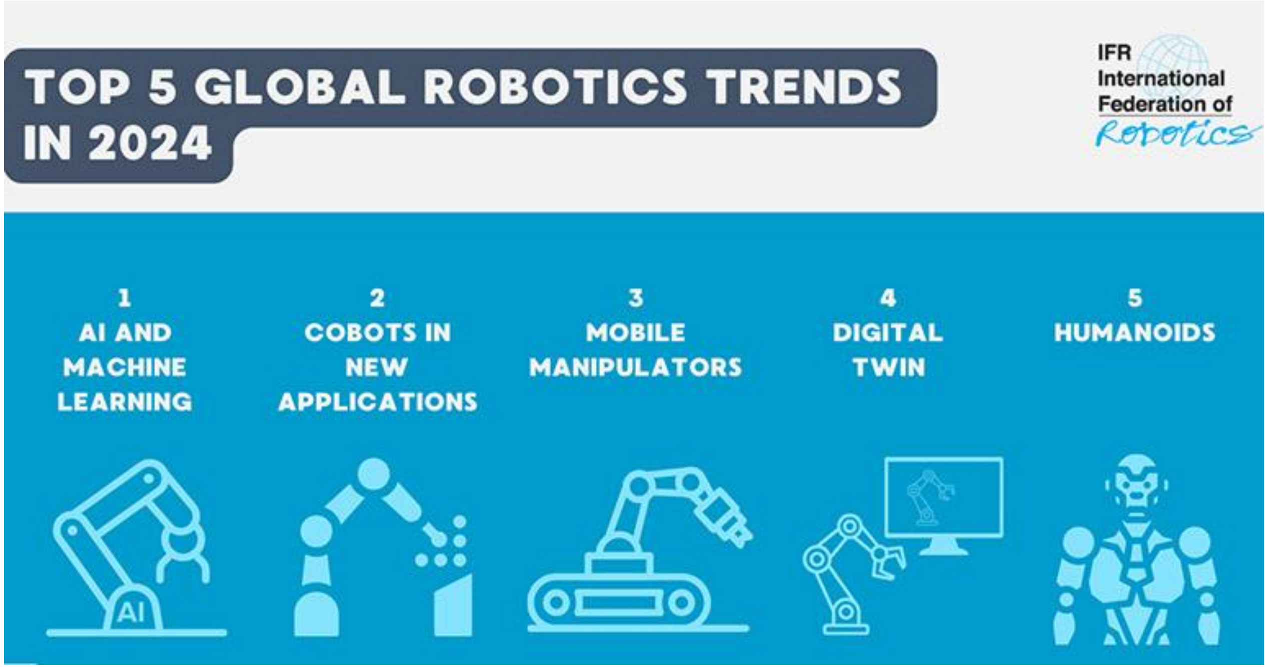






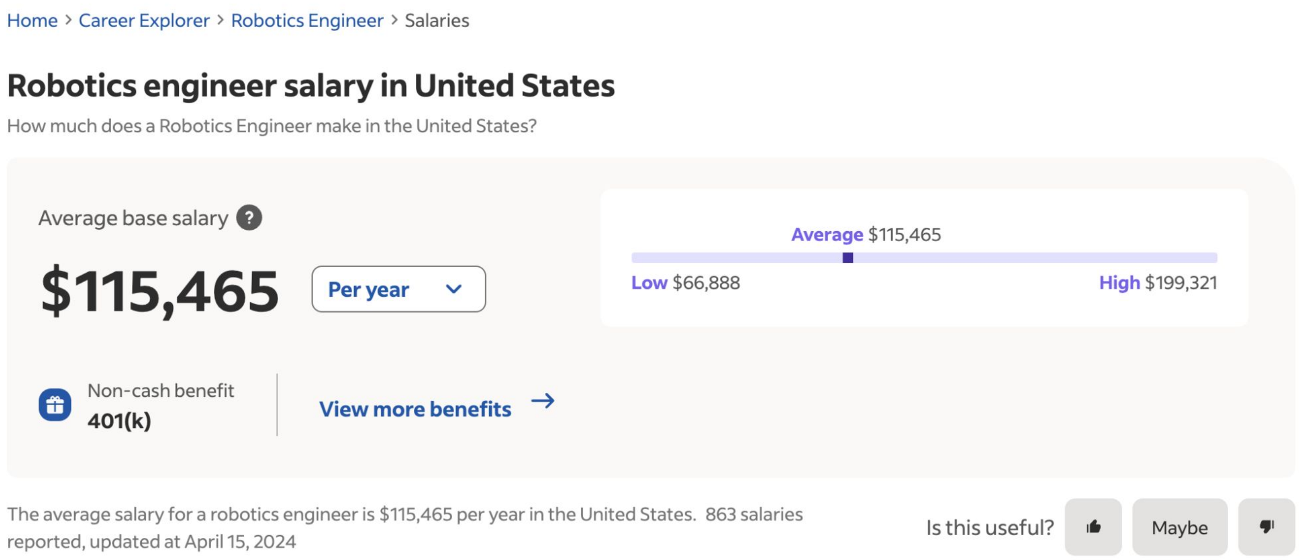
https://www.weforum.org/reports/the-future-of-jobs-report-2020/in-full/infographics-e4e69e4de7

# Trends in robotics in 2024




https://www.therobotreport.com/top-5-robotics-trends-for-2024-according-to-the-ifr/

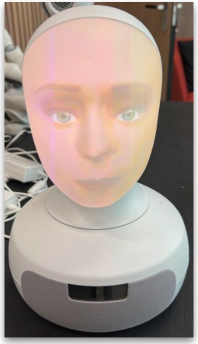
# It is perfect time to be robotics developer/engineer




https://www.indeed.com/career/robotics-engineer/salaries

With the HCI and Autonomous Systems curriculum, you already have a good starting point. And if you **extend your knowledge for 1-2 more years** by conducting research in robotics, you will have a **competitive advantage** in academia and industry.

# Robots from AIR-Lab

# General info and a few rules

The exam will be a combi exam, so you can expect approx. 40 questions:

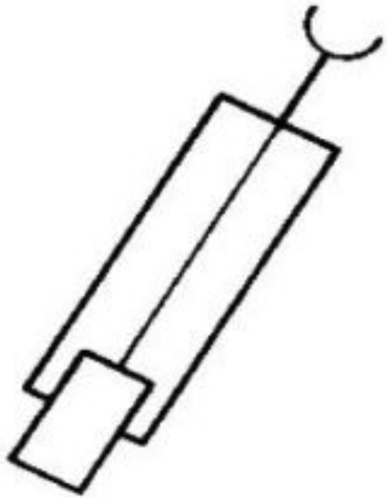
- Open-ended
- Fill in the blanks
- True/False

The exam date is already announced please check MyTimeTable/Emails/Canvas.
Do not forget to register to exam.

Any question about the format of the exam?

# Question 1

Q: Describe the behavior of the Braitenberg Vehicle 1 (see figure right) in an open-ended environment where temperature varies. Recall that this vehicle has one temperature sensor directly connected to the actuator.



[for this question

- please assume the temperature can not be negative
- two consecutive readings will not be the same]

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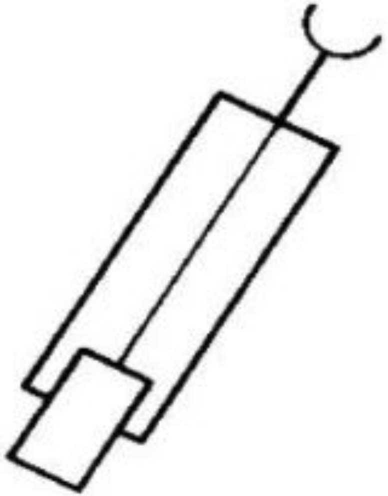

[for this question

- please assume the temperature can not be negative
- two consecutive readings will not be the same a]

Since the sensory reading is directly proportional to its velocity, the vehicle will slow down in cold areas, and speed up in hot areas.

(If you assume the temperature value will never be the same value for two consecutive sensory readings,  it will never stop)

# Question 2

Mark below sentences as True or False.

a) A proprioceptive sensor measures the quantities related to internal state of the agent.
b) Cameras located on robot’s forehead are types of proprioceptive sensors.

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A) True
B) False

# Question 3

In the image above, you see two types of robots (Affetto (left) and soft walking robot (right)). Considering the robots in the images, mark the following sentences as True or False.

A) Both robots are examples of embodied agents.
B) Soft walking robot (right) has the morphology of a human hand.

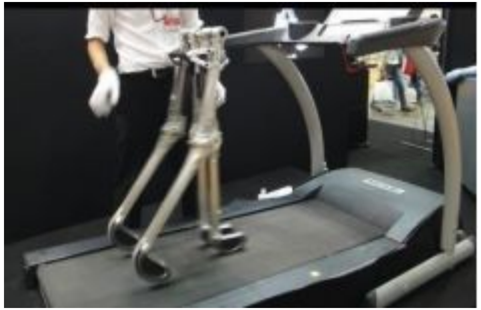
# Question 2

In the image above, you see two types of robots (Affetto (left) and soft walking robot (right). Considering the robots in the image, mark the following sentences as True or False.

A) Both robots are examples of embodied agents.
B) Soft walking robot (right) has the morphology of a human hand.

A) True, B) False

# Question 4

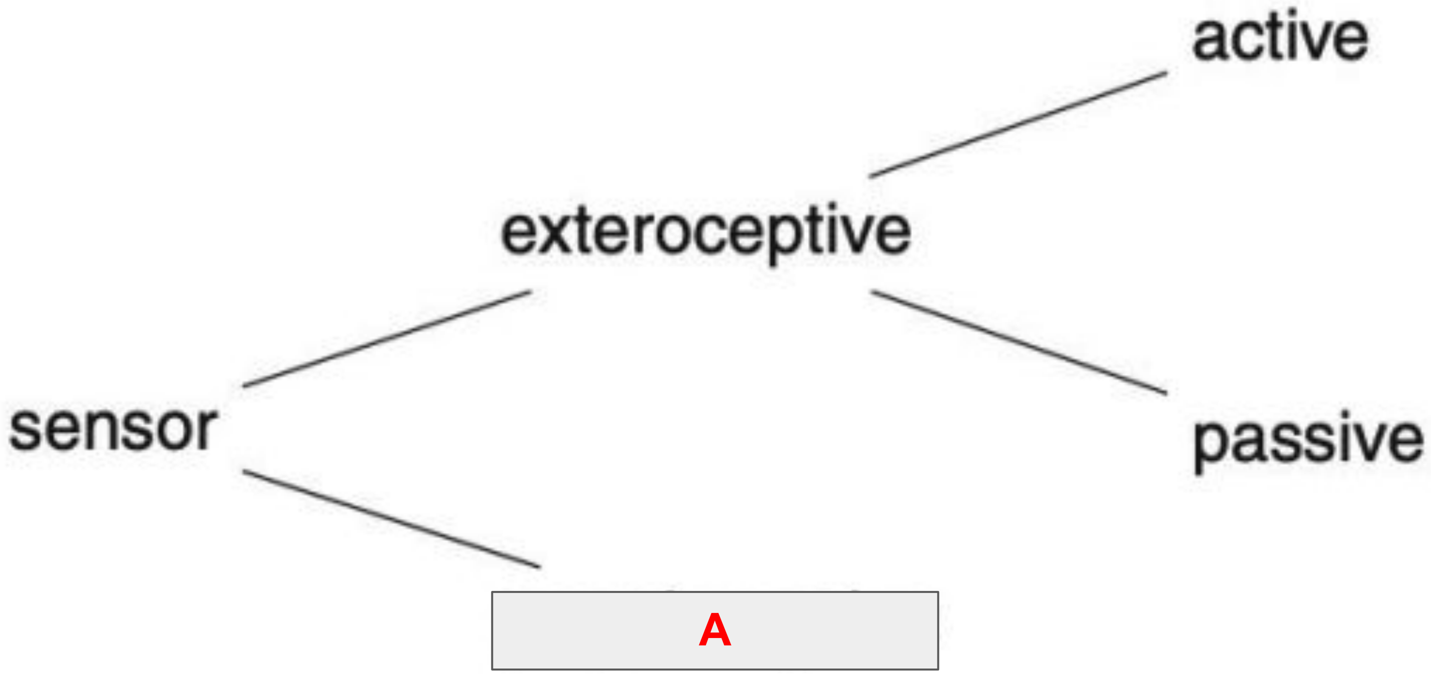


Why passive walkers are considered as a good example for Morphological computation? Please write a few sentences to answer this questions.

# Question 5

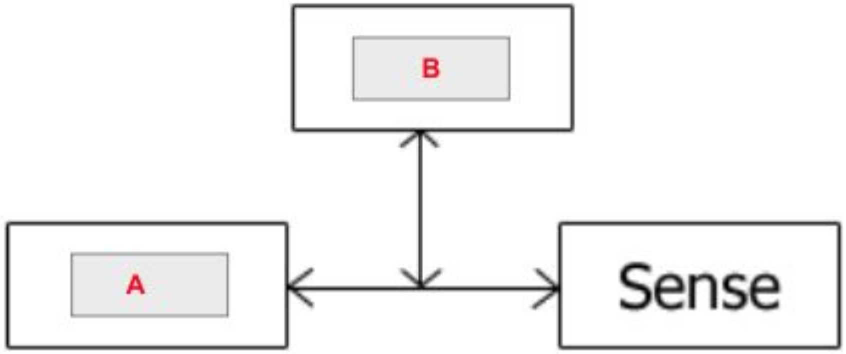
Why industrial robots that can perform precise computation for pick and place tasks can not be considered agents that perform Morphological computation? Please write a few sentences to answer this question.

# Question 6



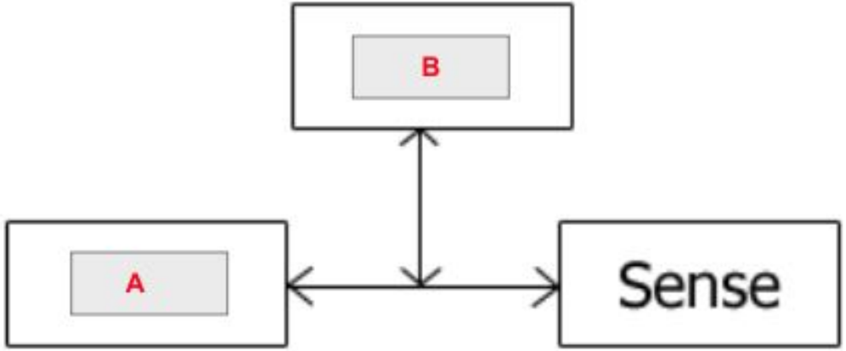

In the lectures, we used the figure above to classify sensor types. Write the sensory type of the Box A (shown in red in the above figure) and define passive sensors.

# Question 7




The hybrid deliberate/reactive paradigm can be visualized in the figure above. Write the functions that are represented with the boxes A and B.

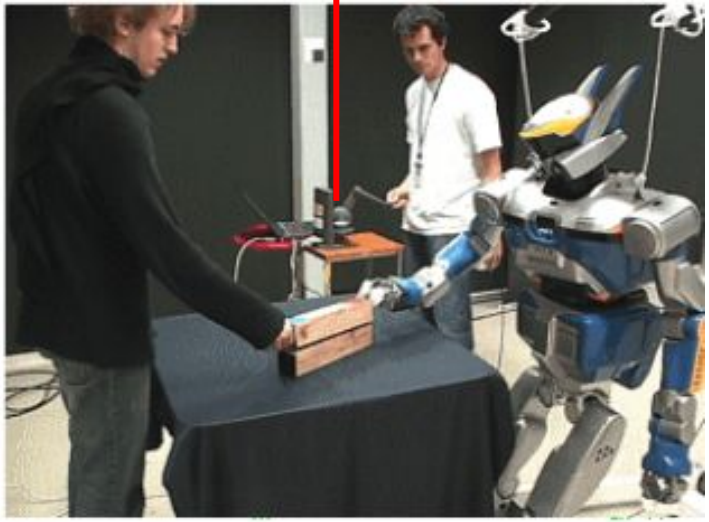
# Question 7




The hybrid deliberate/reactive paradigm can be visualized in the figure above. Write the functions that are represented with the boxes A and B.

A) Act
B) Plan

# Question 8

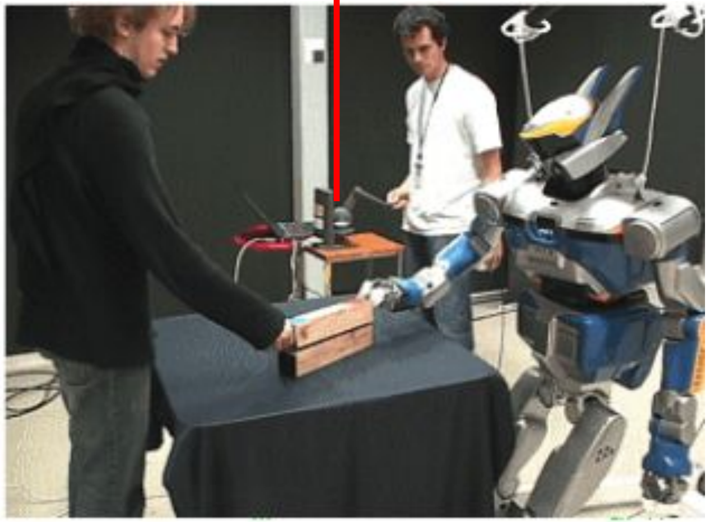



In the figure above, the researchers designed an experiment where an operator controls the robot via a haptic device.

What is the type of learning from demonstration addressed in this experiment?

# Question 8

Haptic device



Teleoperation

In the figure above, the researchers designed an experiment where an operator controls the robot via a haptic device.
What is the type of learning from demonstration addressed in this experiment?

# Question 9

Fill in the blanks for the sentences related to Q-learning below:

A) In reinforcement learning, the 1) ________________ is the term used to represent the numerical feedback that indicates the success or desirability of an action and/or state.

B) During the RL experiment the agent updates its 2)________________ based on the observed rewards and the estimated future rewards.

# Question 9

Fill in the blanks for the sentences related to Q-learning below:

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B) During the RL experiment the agent updates its 2)________________ based on the observed rewards and the estimated future rewards.

Answer: 1) reward, 2) Q-Value, Q-functions, Q-table

# Question 10

What is the role of the controller in autonomous systems? Please frame your answer in the context of the action-perception cycle.

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What is the role of the controller in autonomous systems? Please frame your answer in the context of the action-perception cycle.

A: The brain of the robot regulates the state of the actuator based on the current and previous states of the sensors.

# Question 11

Give the names of two electronic circuit components that you used for your hardware project and describe their functions. Note that you can also give use cases for their functions.

# Question 12

Give the names of two electronic circuit components that you used for your hardware project and describe their functions. Note that you can also give use cases for their functions

A:
Any two electronic components will be accepted, and you do not need to be precise while describing their functions. For example, you can say we used LEDs to debug the circuits or we used cameras to get images of the environment.

Good luck with your exams.